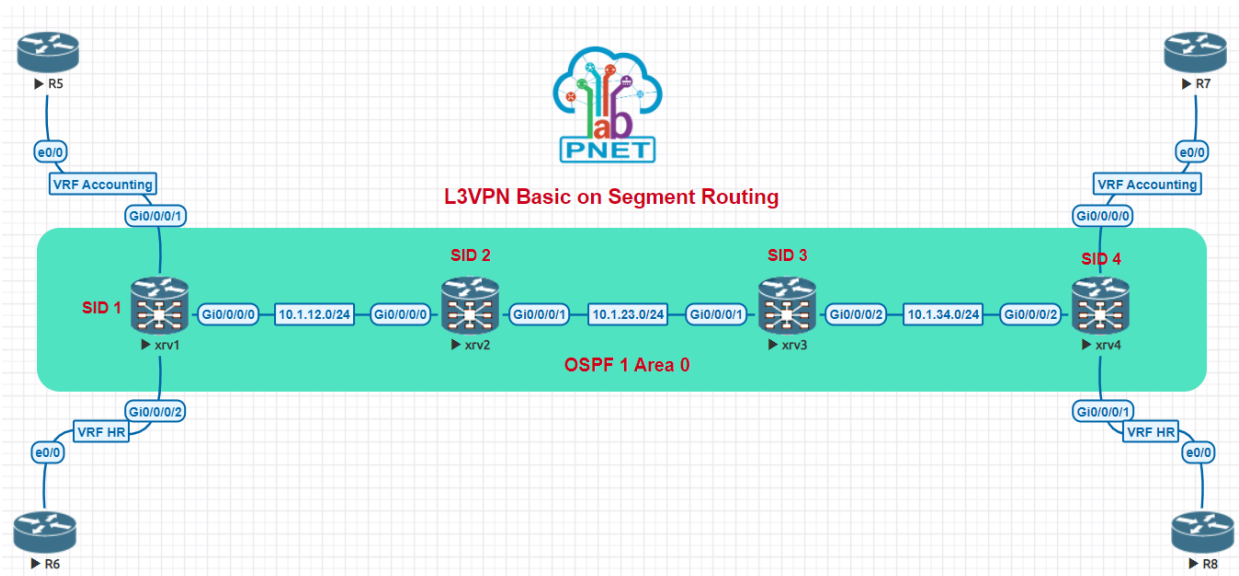


L3VPN BASIC ON SEGMENT ROUTING

Lab Topology:

Please use the following topology to complete this lab exercise:



Lab Objective:

The objective of this lab exercise is for you to learn and understand how can you configure L3VPN on Segment Routing.

Task list:

Task 1: Configure the hostname and IP address for the interface for Routers as topo.

Task 2: Config Segment Routing Global Block for each Router as following table :

Router	Segment Routing Global Block
R1	16000 – 23999
R2	24000 – 31999
R3	32000 – 39999
R4	40000 - 47999

Task 3: Config OSPF Routing in 4 Routers and enable Segment Routing on the area of OSPF.

Task 4: Config VRF in 2 PE Router (XRV1 & XRV4).

Task 5: Config BGP VPNv4 between 2 PE Router (XRV1 & XRV4). Advertise the network on VRF to BGP.



Task 6: Config IP, Static Route for CE Routers (R5,R6,R7,R8) and check the connection between CE Routers.

SOLUTION:

Task 1: Configure the hostname and IP address for the interface for Routers as topo.

On R1

```
!  
hostname R1  
interface Loopback0  
  ipv4 address 1.1.1.1 255.255.255.255  
!  
interface GigabitEthernet0/0/0/0  
  ipv4 address 10.1.12.1 255.255.255.0  
!
```

On R2

```
!  
hostname R2  
interface Loopback0  
  ipv4 address 2.2.2.2 255.255.255.255  
!  
interface GigabitEthernet0/0/0/0  
  ipv4 address 10.1.12.2 255.255.255.0  
!  
interface GigabitEthernet0/0/0/1  
  ipv4 address 10.1.23.2 255.255.255.0  
!
```

On R3

```
!  
hostname R3  
interface Loopback0  
  ipv4 address 3.3.3.3 255.255.255.255  
!  
interface GigabitEthernet0/0/0/1  
  ipv4 address 10.1.23.3 255.255.255.0  
!  
interface GigabitEthernet0/0/0/2  
  ipv4 address 10.1.34.3 255.255.255.0  
!
```

On R4

```
!  
hostname R4  
interface Loopback0
```





```
ipv4 address 4.4.4.4 255.255.255.255
!
interface GigabitEthernet0/0/0/2
ipv4 address 10.1.34.4 255.255.255.0
!
```

Task 2: Config Segment Routing Global Block for each Router as following table :

On R1

```
!
segment-routing
global-block 16000 23999
!
```

On R2

```
!
segment-routing
global-block 24000 31999
!
```

On R3

```
!
segment-routing
global-block 32000 39999
!
```

On R4

```
!
segment-routing
global-block 40000 47999
!
```

Task 3: Config OSPF Routing in 4 Routers and enable Segment Routing on the area of OSPF.

On R1

```
!
router ospf 1
router-id 1.1.1.1
segment-routing mpls
segment-routing forwarding mpls
area 0
interface Loopback0
passive enable
prefix-sid index 1
!
interface GigabitEthernet0/0/0/0
network point-to-point
!
!
```

On R2





```
!  
router ospf 1  
  segment-routing mpls  
  segment-routing forwarding mpls  
  area 0  
    interface Loopback0  
      passive enable  
      prefix-sid index 2  
    !  
    interface GigabitEthernet0/0/0/0  
      network point-to-point  
    !  
    interface GigabitEthernet0/0/0/1  
      network point-to-point  
    !  
  !  
!
```

On R3

```
!  
router ospf 1  
  segment-routing mpls  
  segment-routing forwarding mpls  
  area 0  
    interface Loopback0  
      passive enable  
      prefix-sid index 3  
    !  
    interface GigabitEthernet0/0/0/1  
      network point-to-point  
    !  
    interface GigabitEthernet0/0/0/2  
      network point-to-point  
    !  
  !  
!
```

On R4

```
!  
router ospf 1  
  segment-routing mpls  
  segment-routing forwarding mpls  
  area 0  
    interface Loopback0  
      passive enable  
      prefix-sid index 4  
    !  
    interface GigabitEthernet0/0/0/2
```





```
network point-to-point
!
!
!
```

Task 4: Config VRF in 2 PE Router (XRV1 & XRV4).

On R1,R4

```
!
vrf HR
address-family ipv4 unicast
import route-target
1:200
!
export route-target
1:200
!
!
!
vrf Accounting
address-family ipv4 unicast
import route-target
1:100
!
export route-target
1:100
!
!
!
```

Task 5: Config BGP VPNv4 between 2 PE Router (XRV1 & XRV4). Advertise the network on VRF to BGP.

On R1

```
!
router bgp 12345
bgp router-id 1.1.1.1
address-family vpnv4 unicast
!
neighbor-group L3VPN
remote-as 12345
update-source Loopback0
address-family vpnv4 unicast
!
!
```





```
neighbor 4.4.4.4
  use neighbor-group L3VPN
!
vrf HR
  rd 2:2
  address-family ipv4 unicast
    network 192.168.16.0/24
  !
!
vrf Accounting
  rd 1:1
  address-family ipv4 unicast
    network 192.168.15.0/24
  !
!
!
```

On R4

```
!
router bgp 12345
  bgp router-id 4.4.4.4
  address-family vpnv4 unicast
  !
  neighbor-group L3VPN
  remote-as 12345
  update-source Loopback0
  address-family vpnv4 unicast
  !
!
neighbor 1.1.1.1
  use neighbor-group L3VPN
!
vrf HR
  rd auto
  address-family ipv4 unicast
    network 192.168.48.0/24
  !
!
vrf Accounting
  rd auto
  address-family ipv4 unicast
    network 192.168.47.0/24
  !
!
!
```





Task 6: Config IP, Static Route for CE Routers (R5,R6,R7,R8) and check the connection between CE Routers.

On R5

```
!  
interface Ethernet0/0  
 ip address 192.168.15.5 255.255.255.0  
!  
ip route 0.0.0.0 0.0.0.0 192.168.15.1  
!
```

On R6

```
!  
interface Ethernet0/0  
 ip address 192.168.16.6 255.255.255.0  
!  
ip route 0.0.0.0 0.0.0.0 192.168.16.1  
!
```

On R7

```
!  
interface Ethernet0/0  
 ip address 192.168.47.7 255.255.255.0  
!  
ip route 0.0.0.0 0.0.0.0 192.168.47.4  
!
```

On R8

```
!  
interface Ethernet0/0  
 ip address 192.168.48.8 255.255.255.0  
!  
ip route 0.0.0.0 0.0.0.0 192.168.48.4  
!
```

Verification:

R5#**ping 192.168.47.7**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.47.7, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 16/24/45 ms

R6#**ping 192.168.48.8**

Type escape sequence to abort.





Sending 5, 100-byte ICMP Echos to 192.168.48.8, timeout is 2 seconds:
!!!!!!

//2 Pairs CE belong to 2 VRF could reach to each other. Let's check what's happen on Core Area.

RP/0/0/CPU0:XR1#show mpls forwarding

Mon Oct 12 15:06:16.211 UTC

Local Bytes	Outgoing Label	Prefix or ID	Outgoing Interface	Next Hop	
Switched	Label	or ID	Interface		
-----	-----	-----	-----	-----	-----
16002	Pop	SR Pfx (idx 2)	Gi0/0/0/0	10.1.12.2	0
16003	24003	SR Pfx (idx 3)	Gi0/0/0/0	10.1.12.2	0
16004	24004	SR Pfx (idx 4)	Gi0/0/0/0	10.1.12.2	0
512127					
24000	Pop	SR Adj (idx 0)	Gi0/0/0/0	10.1.12.2	0
24001	Pop	SR Adj (idx 0)	Gi0/0/0/0	10.1.12.2	0
24002	Aggregate	Accounting: Per-VRF Aggr[V]	\	Accounting	520
24003	Aggregate	HR: Per-VRF Aggr[V]	\	HR	1040

//2 Labels for L3VPN.

RP/0/0/CPU0:XR1#show bgp vpnv4 unicast

Mon Oct 12 15:12:36.645 UTC

BGP router identifier 1.1.1.1, local AS number 12345

BGP generic scan interval 60 secs

Non-stop routing is enabled

BGP table state: Active

Table ID: 0x0 RD version: 0

BGP main routing table version 15

BGP NSR Initial initsync version 7 (Reached)

BGP NSR/ISSU Sync-Group versions 0/0

BGP scan interval 60 secs

Status codes: s suppressed, d damped, h history, * valid, > best
i - internal, r RIB-failure, S stale, N Nexthop-discard

Origin codes: i - IGP, e - EGP, ? - incomplete

Network Next Hop Metric LocPrf Weight Path

Route Distinguisher: 1:1 (default for vrf Accounting)

*> 192.168.15.0/24 0.0.0.0 0 32768 i





```
*>i192.168.47.0/24 4.4.4.4 0 100 0 i
Route Distinguisher: 2:2 (default for vrf HR)
*> 192.168.16.0/24 0.0.0.0 0 32768 i
*>i192.168.48.0/24 4.4.4.4 0 100 0 i
Route Distinguisher: 4.4.4.4:0
*>i192.168.48.0/24 4.4.4.4 0 100 0 i
Route Distinguisher: 4.4.4.4:1
*>i192.168.47.0/24 4.4.4.4 0 100 0 i
```

//VPNv4 could be advertised between 2 PE.

RP/0/0/CPU0:XR1#show bgp vrf HR 192.168.48.0/24

Mon Oct 12 15:10:42.143 UTC

BGP routing table entry for 192.168.48.0/24, Route Distinguisher: 2:2
Versions:

Process	bRIB/RIB	SendTblVer
Speaker	12	12

Last Modified: Oct 12 14:47:53.143 for 00:22:49

Paths: (1 available, best #1)

Not advertised to any peer

Path #1: Received by speaker 0

Not advertised to any peer

Local

4.4.4.4 (metric 4) from 4.4.4.4 (4.4.4.4)

Received Label 24002

Origin IGP, metric 0, localpref 100, valid, internal, best,
group-best, import-candidate, imported

Received Path ID 0, Local Path ID 1, version 12

Extended community: RT:1:200

Source AFI: VPNv4 Unicast, Source VRF: default, Source Route
Distinguisher: 4.4.4.4:0

RP/0/0/CPU0:XR1#show bgp vrf Accounting 192.168.47.0/24

Mon Oct 12 15:11:39.679 UTC

BGP routing table entry for 192.168.47.0/24, Route Distinguisher: 1:1
Versions:

Process	bRIB/RIB	SendTblVer
Speaker	13	13

Last Modified: Oct 12 14:47:53.143 for 00:23:46

Paths: (1 available, best #1)

Not advertised to any peer

Path #1: Received by speaker 0

Not advertised to any peer

Local

4.4.4.4 (metric 4) from 4.4.4.4 (4.4.4.4)





Received Label 24003

Origin IGP, metric 0, localpref 100, valid, internal, best,
group-best, import-candidate, imported
Received Path ID 0, Local Path ID 1, version 13
Extended community: RT:1:100
Source AFI: VPNv4 Unicast, Source VRF: default, Source Route
Distinguisher: 4.4.4.4:1

//Label 24002, 24003 will be use for corresponding VRF's route. Let's use traceroute for check the label on packet.

R5#**traceroute 192.168.47.7**

Type escape sequence to abort.

Tracing the route to 192.168.47.7

VRF info: (vrf in name/id, vrf out name/id)

1	192.168.15.1	29 msec	2 msec	2 msec	
2	10.1.12.2	[MPLS: Labels 24004/24003 Exp 0]	39 msec	22 msec	35 msec
3	10.1.23.3	[MPLS: Labels 32004/24003 Exp 0]	34 msec	18 msec	30 msec
4	10.1.34.4	31 msec	31 msec	26 msec	
5	192.168.47.7	16 msec	21 msec	*	

//It's very similar with L3VPN over MPLS.

